



Financial Institutions & Markets- ENT 309

PPT #8, IITD, Summer, 2026

Agenda

1

CENTRAL BANKS & THEIR ACTIVITIES

- RBI – Role & Objectives
- Monetary Policy Framework
- Monetary Policy Instruments
- Monetary Policy Implementation
- Monetary Policy Transmission
- FX Intervention by RBI
- Financial Stability
- Commercial Banking Regulations
- Payment & Settlement Systems
- Financial Inclusion

Central Banks



- India The Reserve Bank of India
- US Federal Reserve System
 - Federal Board of Governors & 12 regional Federal Reserve Banks (Regulatory & Supervisory responsibilities)
 - Federal Open Market Committee (Interest Rates & Monetary Policy).
- UK Bank of England
- Europe European Central Bank.
 - Consists of 21 member countries who use Euro and whose monetary policy is set up ECB.
 - Many big countries have stayed out. UK, Sweden, Swiss etc. Have their own monetary policy & currency.
- Japan Bank of Japan
- China The People's Bank of China

RBI- Roles & Objectives



RBI- Role & Objectives

- Reserve Bank of India Act 1934
 - "to regulate the issue of banknotes and the keeping of reserves with a view to securing monetary stability in India and to operate the currency and credit system
 - the primary objective of the monetary policy is to maintain price stability while keeping in mind the objective of growth"
- Transact various businesses like
 - Making of short term loans and advances to banks and other institutions,
 - Dealing in Derivatives, purchase and sale of Government Securities,
 - Making & Issuance of Bank notes
 - Act as a Banker for Central & State Govts. Accept deposits from them & make payments
 - Lender of Last Resort



RBI- Role & Objectives

MONETARY POLICY FUNCTIONS

- To maintain a level of inflation, interest rates, economic growth & credit supply.
- **MANAGEMENT OF PUBLIC DEBT OF CENTRAL & STATE GOVT**
 - Powers given under Government Securities (GS) Act 2006
 - Act prescribes the procedures and modalities for this purpose.
- **FOREIGN EXCHANGE MANAGEMENT**
 - Powers given under Foreign Exchange Management Act, 1999 (FEMA)
 - RBI authorizes a person to deal in FX

RBI- Role & Objectives

- **BANKING REGULATION & SUPERVISION**
 - Under Banking Supervision & Regulation Act. 1949 (BR Act)
 - Formulate banking policy, regulate & supervise.
 - Issue and cancel license to banks/NBFCs. Issue directions.
 - RBI can supersede the board of banking companies
- **REGULATION OF DERIVATIVE & MONEY MARKETS**
 - Determines the policy **relating to interest rates or interest rate products** and give directions in that behalf to agencies dealing with such instruments

RBI- Role & Objectives



- **PAYMENT & SETTLEMENT FUNCTION**
 - Under Payment & Settlement Systems Act, 2007 (PSS Act)
 - Power to determine standards for the functioning of payment systems, power to enter and inspect payment systems, to carry out audit and inspections, to issue directions
- **REGULATION OF CREDIT INFORMATION COMPANIES**
 - Credit Information Companies (Regulation) Act, 2005
- **CUSTOMER PROTECTION & FINANCIAL EDUCATION**
 - Make regulations for this purpose.
 - Ombudsman. Customers can file a complaint with RBI

Monetary Policy Framework

Monetary Policy



- Use of Monetary Instruments available with RBI to **influence interest rates, money supply and availability of credit** (loans) etc. to achieve the objectives of Policy.
- Objectives: To maintain a good balance between
 - **Price Stability i.e. maintain inflation in a target range**
 - Economic Growth
 - Financial Stability
- Inflation Targeting is the goal of most Central Banks.

MAIN OBJECTIVE





RBI- Role & Objectives

MONETARY POLICY FUNCTIONS

- Central Government, in consultation with the RBI shall determine the **inflation target in terms of the Consumer Price Index**, once in every five years
- **Monetary Policy Committee (MPC)**: Constituted by Central Govt.
 - Duty is to determine interest rates for meeting inflation target
 - Decision of MPC is binding on RBI
 - 6 members. RBI Guv, Dy Guv of Monetary Policy, One RBI Officer, & 3 persons appointed by Govt.
 - Meets at least 4 times a year
 - Details of MPC discussions shared with public



A Bit of Banking

- What is a Bank Run?
- Banking business is inherently unstable. Why?
 - Bank has deposits which can be withdrawn by the account holders whenever they wish. But the bank cannot recall loans when it wants.
- Bank deposits in India are insured. Up to INR 5 lacs per person per bank.
- Banks create money. What is Fractional Reserve?
 - When a bank gives a loan, it creates new money.
 - Bank keeps only a 'fraction' of its deposits as reserve. Rest used to give loans.
- **Money Multiplier:** This process of deposit followed by lending is repeated many times. Thus INR One injected by RBI in the system will create Rs 6.21 (money multiplier as of Dec 2025).

Banking: CRR, SLR



- CENTRAL BANK AS THE LENDER OF LAST RESORT
- Silicon Valley Bank Failure Explainer
- My Article: Failure of Silicon Valley Bank

What reserves are banks required to keep in India?

- **Cash Reserve Ratio CRR:** 3% of the deposits with the bank are to be kept in cash with RBI. End of every day. No interest is paid on CRR.
- **Statutory Liquidity Ratio SLR:** 18% of the deposits to be invested in Govt Approved Securities, Gold or cash. which can be quickly liquidated.
- Why is CRR & SLR needed?
- **CRR Calculated & Met Daily:** Imbalance in deposits & withdrawals.

Call Money Market



FOR MEETING CRR DAILY

- **Call Money Market:** One option for Banks is to borrow OVERNIGHT from other banks. This is the **Call Money Market**.
- Borrowing is done without any collateral
- Banks and Primary Dealers (PD: a special category of financial firms)
- Insurance companies/ Mutual Funds participate only as Lenders through PDs.
- **Weighted Average Call Rate (WACR)** of overnight trxn in Call Money Market is an important indicator of liquidity and interest rates in the market.

Monetary Policy Implementation



- **FLEXIBLE INFLATION TARGETING :**
 - Inflation target defined in terms of Consumer Price Index (CPI)
 - Will be set by GoI + RBI once every 5 years.
 - Manage interest rates to meet inflation, growth, credit targets.
 - How to do it? Goal is to track & align **Weighted Average Call Rate (WACR) close** to Repo rate.
 - Manage interest rate movement within an upper and lower range

Monetary Instruments



IMPLEMENTING MONETARY POLICY: LAF(LIQUIDITY ADJUSTMENT FACILITY)

- **Repo Transaction:** RBI provides loans (liquidity) to banks against collateral of Govt & other securities. Interest rate for this txn – Repo rate. Decided by MPC
- **Repo: Repurchase Option.** You deposit collateral and then repurchase it.
- **Reverse Repo Transaction :** RBI absorbs liquidity i.e. takes deposits from banks against collateral. Interest rate for this txn- Reverse Repo Rate
- **Repo/ Reverse Repo Maturity:** Is from Overnight (single day maturity) up to 14 days. Sold through auctions.
- **LTRO- Long Term Repo Operation.** Maturity from 14 days up to a few years
- **VARIABLE rate repo** auctions are also done depending on the liquidity needs of the banking system.

Monetary Instruments



INSTRUMENTS WITH RBI FOR IMPLEMENTING MONETARY POLICY

- **Standard Deposit Facility SDF:** Banks can deposit money with RBI at their discretion. No collateral involved. Rate of Interest to be decided by RBI.
- Repo, Reverse Repo, LTRO, Variable Rate Repo & SDF – all these instruments constitute LIQUIDITY ADJUSTMENT FACILITY (LAF)
- There is a limit for each bank on how much it can borrow using LAF.
- **MARGINAL STANDING FACILITY (MSF):** Allows Banks to borrow over and above what is available through LAF. To meet emergency liquidity requirements of banks. Strictly overnight. Available on demand. Higher interest rate than Repo – as a penalty. Against collateral of excess SLR holdings.

Monetary Instruments



INTEREST RATES OF THESE PRODUCTS

- Repo & Reverse Repo rate decided by the MPC
- MSF, SDF rates are decided by RBI from time to time.
- **Current Rates:** <https://www.rbi.org.in/#mainsection>
 - Repo (5.25%), MSF (5.5%)
 - SDF (5%), Reverse Repo (3.35%)
 - WACR (4.2% to 5.5%) on June 23, 2026

Monetary Policy Implementation



MONETARY POLICY IMPLEMENTATION USING DIFFERENT PRODUCTS

- For short-term change in liquidity LIQUIDITY ADJUSTMENT FACILITY (LAF)
 - **Open Market Operations (OMO):** RBI has the discretion to conduct overnight/ longer term, repo/ reverse repo auctions at variable/fixed rates. To inject or absorb liquidity.
 - SDF
 - MSF
- For long term change in Liquidity
 - **Long Term Variable Rate Repo & Reverse Repo (LTRO/ LTRRO)**
 - **Cash Reserve Ratio:**
 - An increase in CRR sucks out money from circulation. A decrease in CRR increase liquidity in the system.

EXAMPLE



Each bank has a limit on how much Repo transactions it can do with RBI. Take a case when the bank has exhausted its Repo txn limit.

XYZ bank needs to borrow more overnight Rupee funds.

- Let us say it tries to borrow from other banks.
- If there is system level low liquidity or availability of money, overnight lending rates will go up.
- But XYZ Bank also has an option to borrow from RBI under MARGINAL STANDING FACILITY (MSF) at **MSF INTEREST RATE**. Borrowing under MSF is over and above the bank specific limit under Repo Transaction.

Hence MSF Interest Rate acts as the Ceiling Rate/ Upper Cap for overnight borrowing.

EXAMPLE (Contd.)



Similarly, the bank might want to deposit funds with RBI beyond what it can deposit using Reverse Repo Transaction.

The bank has the option of using STANDING DEPOSIT FACILITY (SDF). Under SDF, banks can deposit money without receiving any collateral from RBI and earn interest.

XYZ bank needs to lend overnight Rupee funds.

- If there is too much supply of money, interest rate in money market will fall.
- The bank can deposit with RBI under SDF (without collateral) which is at a rate higher than reverse repo rate.
- So, SDF rate effectively acts as a Floor Rate for overnight interest rates.



EXAMPLE (Contd.)

The range between interest rate ceiling (MSF rate) and floor (SDF rate) is called the INTEREST RATE CORRIDOR OR LIQUIDITY ADJUSTMENT FACILITY (LAF) CORRIDOR

If WACR is close to SDF rate

- Excess liquidity in the system
- RBI can do a **variable rate reverse repo (VRRR)** auction to bring the WACR rate closer to repo rate. On Apr 17, 2026, RBI did such an auction at 5.24% rate
- OMO sale of govt. securities is used to suck out liquidity over a longer term

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THE HINDU **RBI pulls out ₹2 lakh crore transient liquidity via 7-day VRRR auction**

The central bank had received higher bids worth ₹2,28,098 crore than the notified amount of ₹2 lakh crore in the auction. The RBI accepted the bid at a 5.24% cut-off rate and 5.23% weighted average rate



EXAMPLE (Contd.)

- LAF Corridor: Between Interest Rate for MSF (ceiling) & SDF (floor)

If WACR is close to MSF rate

- Less liquidity in the market (liquidity under strain)
- RBI can do additional variable rate repo (VRR) auction to bring WACR closer to repo rate. Done on July 8, 2026 at 5.26%
- Or it can OMO purchase of govt. securities to increase liquidity for longer term



Result of the Overnight Variable Rate Repo (VRR) auction held on July 08, 2026

Tenor	1-day
Notified Amount (in ₹ crore)	25,000
Total amount of bids received (in ₹ crore)	665
Amount allotted (in ₹ crore)	665
Cut off Rate (%)	5.26
Weighted Average Rate (%)	5.26
Partial Allotment Percentage of bids received at cut off rate (%)	NA

Monetary Policy Transmission

Monetary Policy Transmission



- How do changes in Monetary Policy (change in interest rates, CRR etc.) transmit to money market interest rates, bond yields, bank deposit and lending rates and asset prices such as stock prices and house prices?
- Economic agents- households, firms and the government- respond to these interest rate changes by adjusting their spending behaviour.
- Example
 - Let us say Interest Rates increase due to monetary policy actions. Loans charge a higher interest, savings deposits pay higher interest. Result: People save more and spend less. Vice versa when interest rates decrease.
 - Change in demand of households and firms. Production by firms increases or decreases to respond to changes in demand.
 - Objectives of price stability and sustainable growth of the economy are achieved.

Monetary Policy Transmission



- Monetary policy actions are felt with a lag of 2-3 quarters on output & with a lag of 3-4 quarters on inflation, and the impact persists for 8-12 quarters.
- Transmission of change in Interest Rates has been found to be most effective in India
- Banks dominate lending in India. So fast transmission of Policy Actions to the interest rates of loans is important.
- But RBI was often unhappy with the speed of Transmission of policy changes to lending rates.

Monetary Policy Transmission



- How do banks fix interest rates on loans?
 - **Int Rate = Benchmark Interest Rate + Spread.** Qu: What is the benchmark?
- Few examples of Benchmark Rates used earlier:
 - **Prime Lending Rate PLR:** Interest rate offered to the 'best – most creditworthy' customer by the bank.
 - **MCLR (Marginal Cost of Funds Based Lending Rate)-** Included the effect of Marginal cost of funds, Repo rate, Operating costs, CRR cost etc.
- Calculation of Benchmark by banks was not transparent. Varied from bank to bank. Difficult for RBI to say if the rate transmission had happened quickly.
- **Now** Banks have to use **EBLR- External Benchmark Lending Rate-** for their lending rates.
- Examples of EBLR: Repo Rate, 3- month T-bill rate, 6- month T-Bill rate etc. These are easily available in public domain and hence transparent.

Foreign Exchange Market Interventions by RBI

FX Market Operations



THREE MAIN OBJECTIVES OF RBI INTERVENTION IN FX

- Reduce excess volatility in exchange rates
 - To maintain adequate level of foreign exchange reserves
 - To develop a healthy FX markets
-
- Volatility makes international trade & investment decisions difficult.
 - Goal of RBI intervention **is not to** achieve a target exchange rate.
 - FX interventions might impact the INR liquidity in the market. How?
 - RBI might take steps to neutralize the impact: **Sterilization Intervention**.
 - Example: Buy USD, Sell INR in FX market. Remove excess supply of INR by conducting Reverse Repo transaction.

FX Market Instruments



- RBI intervenes using Spot, Forwards, Swaps etc
- Rupee exchange rate determined by Spot market rates
- Forward transactions impact spot & forward premium
- Swap impacts only forward premium. Since no change in net FX flow
- Value of a Swap is derived from Spot Rate & Interest Differential Between Two currencies (discussed in FX classes)

Financial Stability

Financial Stability & Crisis



- Stable and well-functioning financial system. Imp for Eco growth
- Financial Crisis often occurs.
- Spillover due to interlinkages between different types of markets (FX & bond markets, Credit & Real Economy i.e. Good/services) and linkages across different geographies (crisis in US impacting financial markets in India).
- Constant struggle between innovation & regulations, freedom and regulation, integration with global network versus independent policy.
- Financial system is a network. Trouble in any part spreads widely.
- The Lehman Bros Crisis of 2008
- Impact on India?

Financial Stability Structure



- Financial Stability and Development Council (FSDC)
 - Strengthen, Institutionalize mechanism for maintaining financial stability, enhancing inter-regulatory coordination and promoting financial sector development.
- Regulators: Reserve Bank of India (RBI), Securities and Exchange Board of India (SEBI), Insurance Regulatory and Development Authority of India (IRDAI), Pension Fund Regulatory and Development Authority (PFRDA)
- Measure & Monitor Risk in the financial system
- Reduce risk and manage its impact by appropriate policies & interventions

Commercial Bank Regulations

Banking Regulations



We understand your world

Particulars	As at 31.03.2026 Audited
CAPITAL AND LIABILITIES	
Capital	1539.34
Employees stock options outstanding	4545.11
Reserves and surplus	556816.45
Deposits	3105250.48
Borrowings	489394.63
Other liabilities and provisions	207340.31
Total	4364886.32

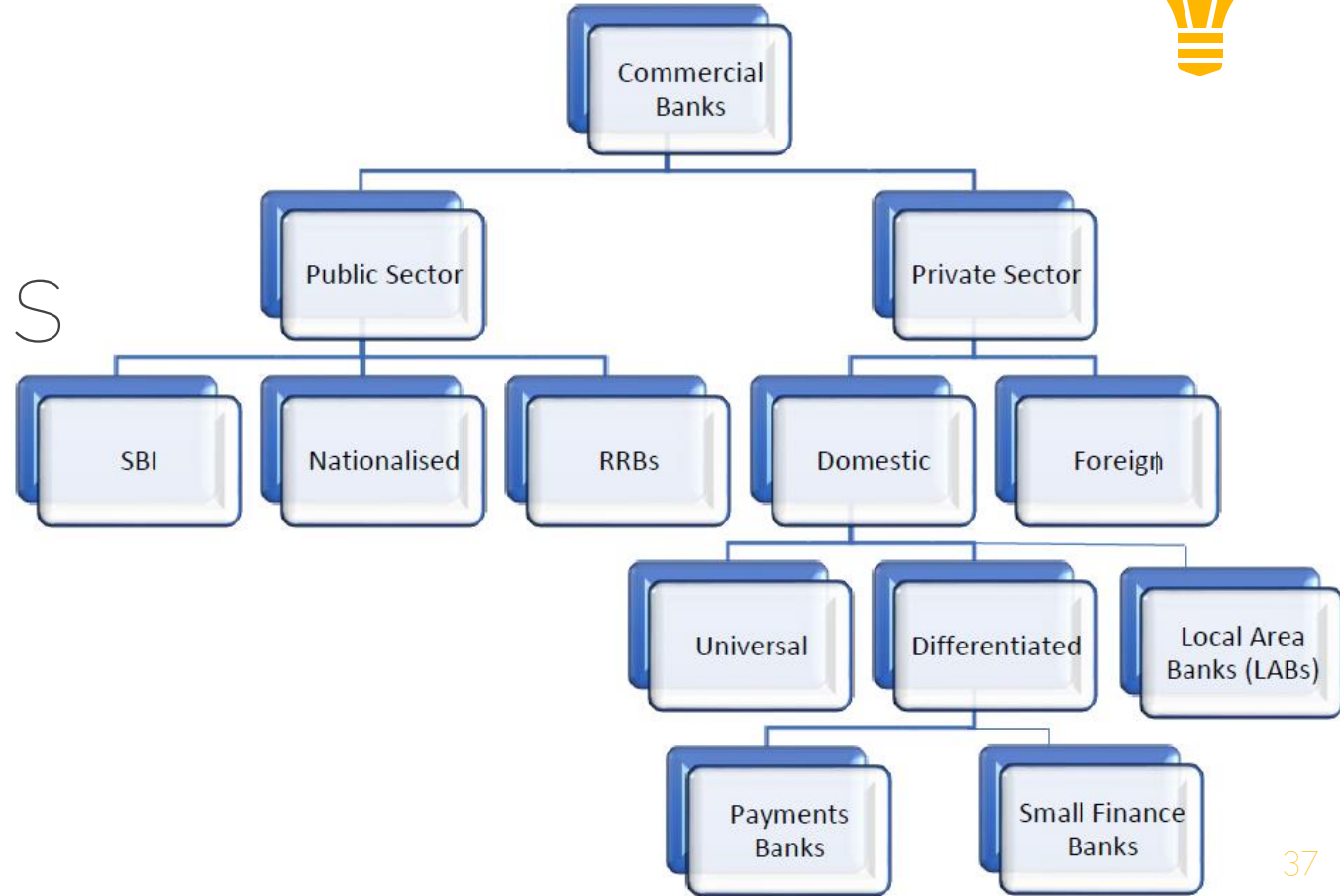
- Banks are highly leveraged. They have relatively low amount of (capital + Reserves & Surplus) and raise a lot of deposits and give a lot of loans.
- Such a set up is prone to instability - High leverage, deposits to be returned on demand, loans given for long tenure (time period)

Banking Regulations



- Video: Why are Strong Banking Regulations Required?
- To encourage people to deposit in banks, deposit insurance is provided
- RBI Lender of Last Resort: To help banks in case of 'a run on the bank' / Liquidity problems.
- Gives rise to 'Moral Hazard' (Change in Behaviour) problem.
 - Since deposit insurance is available, depositors generally don't evaluate the stability of the bank before depositing funds
 - Banks might take higher risk. In case of a problem, expect RBI to help
- Another problem is Information Asymmetry- One party knows more about the risks involved than the counter party
 - Loans: Borrower knows more than the lender about the risks involved
 - Deposits: Bank knows more about its stability & risk position
- Hence well designed Regulatory Framework is necessary to minimize Moral Hazard & Information Asymmetry and provide Stability to the financial system

Indian Banking System Is Diverse





Diff Types of Banks: Why?

How is each of them different?

- To achieve different goals. Directed credit, financial inclusion, level of govt. ownership etc.
- Nationalised Banks: Example Jan Dhan Account opening.
- RRBs: Regional Rural Banks: One per state. To provide savings and loans to economically weaker sections of society in rural and semi-urban areas
- Local Area Banks: Rural, semi-urban. Limited districts.
- Payment Banks: Main goal is to encourage digital payments & small savings accounts
- Small Finance Banks: Main goal is to provide better services to medium & small businesses



Salient Regulations

- Bank Licensing- On Tap now
- Rules related to Branch Expansion: Ex. Foreign banks/ RRBs
- Maintenance of Reserves (CRR & SLR)
- Norms to ensure safety & soundness of banks
 - Related to Income Recognition & Asset (loans) Classification
 - Guidelines on Capital & Liquidity
 - Sufficient capital is required to take care of losses and other shocks
 - Capital Adequacy Norms for lending. Higher the risk category, higher is the capital buffer required (Risk Weighted)



Salient Regulations

- Limit on Exposure to Individual business groups
- Guidelines on how Banks can invest their funds: Approved securities, shares, debentures and bonds, subsidiaries/joint ventures, mutual funds etc.
- Standard Guidelines on How to manage loans which have gone bad (Non Performing Assets). Important so that
 - The correct information about the health of the loan portfolio of the bank is available to everyone
 - The loan pipeline does not get choked due to bad loans
 - Ex: Write off and take loss and continue lending, Sell bad loan portfolio & recover some money from sale etc.



Salient Regulations

- Norms on Risk Management:
 - Credit, interest rate, foreign exchange rate, liquidity, legal, regulatory, reputational, operational risk etc.
 - identify, measure, monitor & control the level of risks undertaken
- Regulate Interest Rates for deposits and loans (EBLR discussed earlier)
 - Interest rates should be Fair & Transparent
- Know Your Customer (KYC) norms: To stop bad actors from misusing the banking system.
 - AML- Anti Money Laundering
 - Sanctions list



Salient Regulations

- Deposit Insurance: To give confidence to depositors.
 - INR 5 lacs per depositor per bank
 - Banks not doing business properly can be shut down by the RBI. How many bank licenses cancelled by RBI from Jan 24 to Dec 25? More than 15.
- Credit Information Companies: To provide information to lenders to do better underwriting of loans.
 - Collect data on all credit given (loans, credit cards etc.) to each individual and current status of repayment from lenders. Once every 15 days.
 - Any lender can ask for this history/information when processing a new loan.
 - 4 credit bureaus in India : CIBIL, CRIF Highmark , Experian & Equifax.

Policies/Intervention Examples

- Decrease loan-to-value ratios:
 - Example – Say, a person buying a home of 1 crore was earlier able to get a loan upto 90 lacs (90%) . With a change in policy, now he will be able to get a loan of only up to 80 lacs (80%). So only a person with at least Rs 20 lacs can now buy a house. This will reduce the number of people who can buy a new home. And reduce growth of home loans.
- Higher/Lower Capital Requirement/ Reserves for lenders:
 - If CRR/SLR is increased, more funds are blocked. And funds available for lending are reduced.
 - Vice Versa

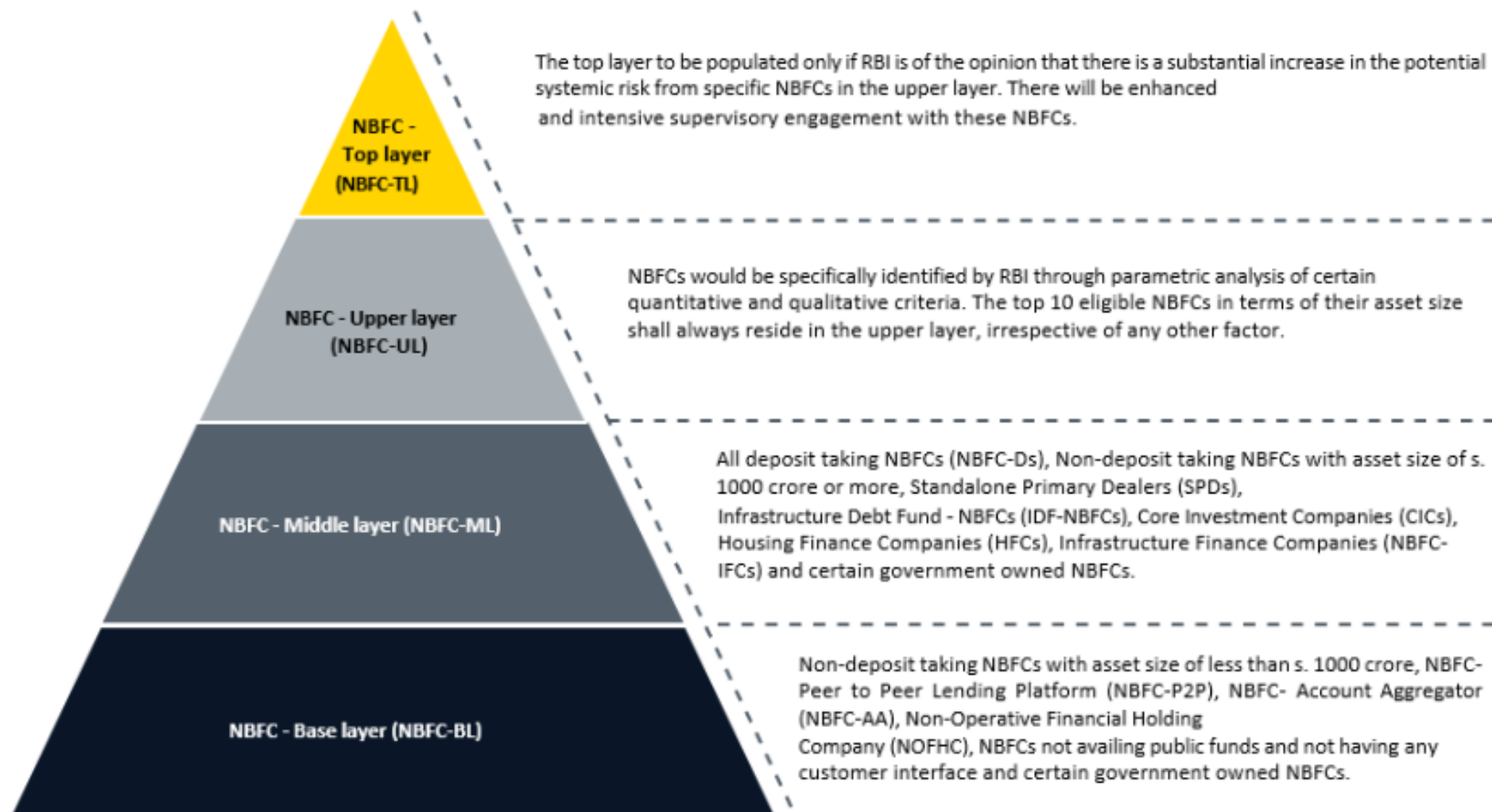
Policies/Intervention Examples

- RBI can put restrictions if credit growth is much higher than economic growth. This can avoid financial instability in future.
- Higher risk weight on certain bank loans. [What are risk weights and how do they work?](#)
- RBI keeps a track of what is happening in the financial market by analyzing data submitted to it regularly by Banks, NBFCs and others.
- RBI conducts regular audits of operations of Banks & NBFCs regularly
- RBI has the powers to issue new regulations as and when required



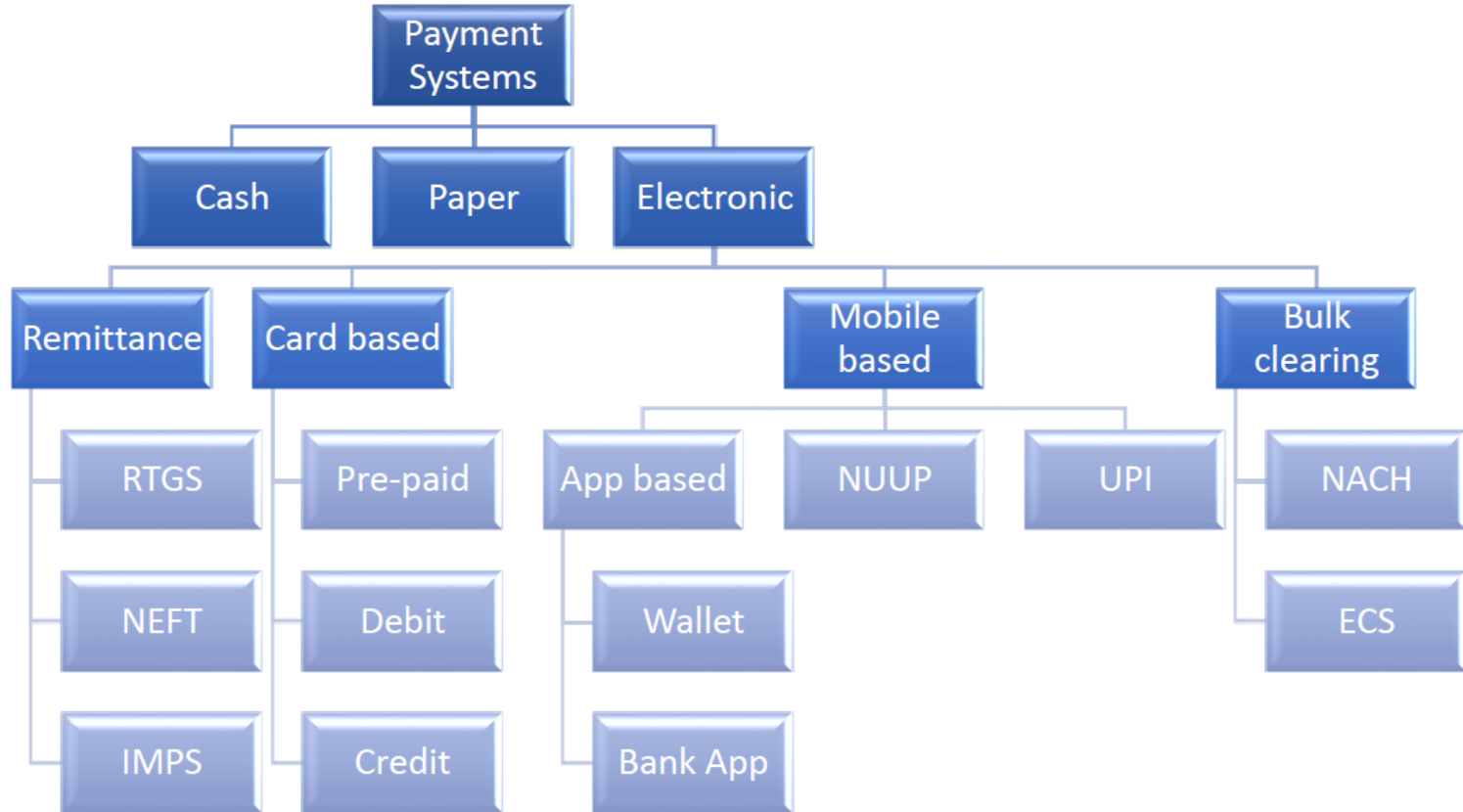
Regulation of NBFCs

- What is an NBFC? Name some big ones
- Types of NBFCs?
- NBFCs play a very important role in increasing availability of loans/credit in the economy.
- Earlier NBFC regulations were not as strict as banks.
- Now the strictness of NBFC regulations is related to the scale or size of its operations. These are called scale based regulations (next slide).



Payment & Settlement System

Retail Payment System



Retail Payment System



- RBI & NPCI (National Payments Corporation of India) manage the payments system in India
- Participants/Networks: Payment system consists of UPI apps (Phone Pe, Google Pe, BHIM), payment aggregators (Razorpay etc), banks, etc.
- Payment Banks play an important role (Airtel Payment Bank, Jio Payment Banks etc)
- Payment Aggregators: Provide multiple services to merchants
- Pre-paid Instruments (example Metro card)
- Tokenization

Bank Payments



- Bank a/c to Bank a/c Transfers using Internet / Mobile Banking
- RBI:
 - RTGS: Uses RBI's Payment System. Real time. Min INR 2 lacs/trxn.
 - NEFT: Thru NEFT processing centre. Batch processing. 30 mins
- NPCI: Instant Payments
 - IMPS: Uses internet banking OR MMID (Mobile Money ID) & mob #
 - UPI: Uses VPA (Virtual Payment Address) & Mobile Apps
- RTGS, NEFT, IMPS, UPI: Available 24X7X365
- <https://www.hdfcbank.com/personal/resources/learning-centre/pay/difference-between-imps-neft-rtgs>



NPCI: Background

- Why was NPCI formed?
 - To develop a national payments framework
 - Initiative of RBI and IBA (Indian Banks Association)
 - Set up in 2008 under PSS Act
- Who owns NPCI?
 - Consortium of 56 banks. 10 core banks.
 - Payment Service Operators.
- Not for profit
- <https://www.npci.org.in/who-we-are/milestone>

NPCI: Products



National Payments Corporation of India

- Rupay: Payment network or switch. Like Visa/Master
- IMPS: Immediate Payment System
- AePS: Aadhar Enabled Payment System. Uses biometric authentication
- UPI
- BBPS: Bharat Bill Payment Service
- NETC: National Electronic Toll Collection. Fastag.
- NFS: National Financial Switch for ATM network

Financial Inclusion

Fin Inclusion: Products



- Savings: Jan Dhan Account
- Lending
 - Priority Sector Lending
 - NBFC-MFI –:specific NBFC license for microfinance
 - Kisan Credit Card (KCC)- Short term loan given to farmers based on their land holding to help in farming.
 - Regional Rural Banks, Small Finance Banks: Specific targets
- PM Jeevan Jyoti Bima Yojana: Life insurance. Rs 2 lacs. Premium Rs. 436 per annum per member.
- PM Suraksha Bima Yojana: Accident Insurance. Up to 2 lacs. Premium Rs. 20 per annum per member
- Atal Pension Yojana: You pay till age of 60 years. Get a pension of 1 k to 5k per month after 60 years till death.
- Business Correspondents: To provide last mile delivery of financial products



Thank You!